

Controlled VLM alignment for grounded reasoning

same budget, same runner, same data caps - only the training signal changes

1 Problem



VLMs can answer correctly while using **weak or wrong visual evidence**

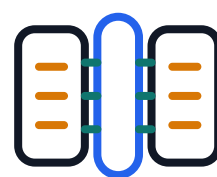
2 Controlled study

Same runner, same data caps, same evaluation path.



Qwen2-VL

QLoRA answer-only, rationale-generative, and explanation-aware runs



BLIP-2

Generative and contrastive-enhanced controls through Q-Former features



Answer-only



Rationale-generative

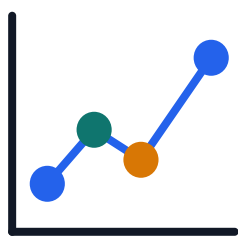


Explanation-aware



Contrastive-enhanced

3 Evaluation



Answer quality

Native dataset metric, not one universal score



Rationale quality

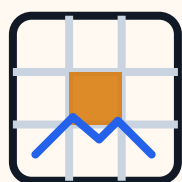
Measured only where gold rationales exist



Visual sensitivity

Masking checks whether answer scores move

4 Impact



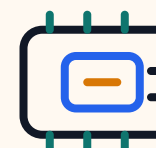
Safer Interpretation

Accuracy, rationale quality and grounding are not merged.



Cleaner claims

Each claim is tied to a control, table, figure, or artifact.



Reusable protocol

The same study can be rerun by a student lab on one GPU.